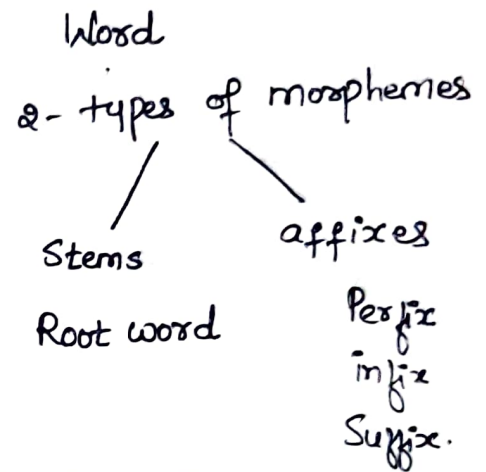
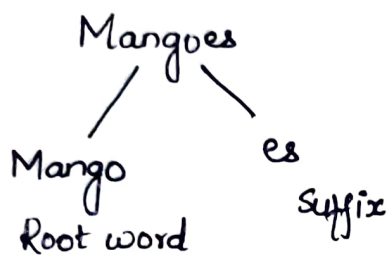


Morphological Parsing:-

- Morphological Parsing is used to identify and find the number of morphemes in a given word.
- Morphemes are the Smallest Indivisible meaningful Units of language which builds a word.

Ex:-



Steps to design Morphological Parser

• Lexicon

- Stores basic Information about a word
- word is Stem or a affix
- if Stem, then whether a Verb stem or Noun stem
- if affix, then whether a Prefix, infix or Suffix.

Morphotactic

Set of Rules to make decisions
• Decides a word appear/not appear before, after or in between other words

Ex:

Use able ness }
 ↓ ↓ ↓
Useableness } Valid
 Obey rules.

able use ness }
 ↓ ↓ ↓
ableuse ness } Invalid
 disobey rules.

Orthographic Rules

Set of Rules used to decide Spelling changes

Ex: Lady + s → Ladys X
Lady + s → Ladies ✓
knife + s → knives ✓

Morphological Analysis:-

Morphology means study of word / making of word

Some words has their own meaning for Ex:

Ex:

Camera Board Pen

Some words are there which when divided into different words, those new words have their own meaning

Ex:

Showcase
/ \
Show Case

Mocktail
/ \
Mock tail

Some words are there which do not have their own meaning but when they are combined with other words, they become meaningful

Ex:-

ing
/ \
Playing killing

FSA for Morphology:-

Word

① Success → (q₀) →^{Stem} (q₁)

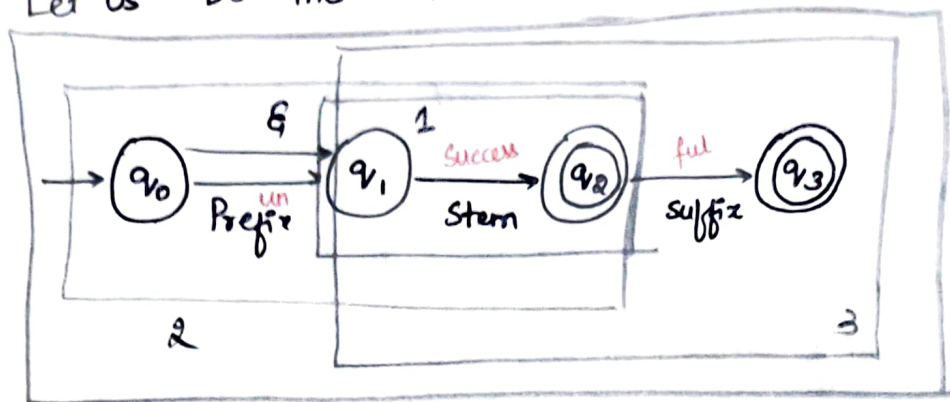
② Unsuccess → (q₀) →^{Prefix} (q₁) →^{Stem} (q₂)

③ Success-ful → (q₀) →^{Stem} (q₁) →^{Suffix} (q₂)

④ Unsuccess-ful → (q₀) →^{Prefix} (q₁) →^{Stem} (q₂) →^{Suffix} (q₃)

this is the kind of DFA.

* Let us Do the NFA :



STEM changes:-

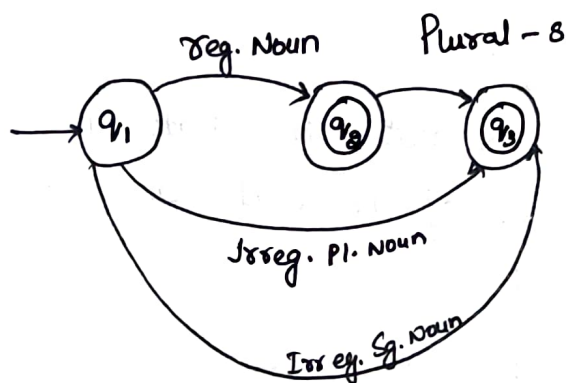
• Some irregular words require stem changes :

• For Ex:

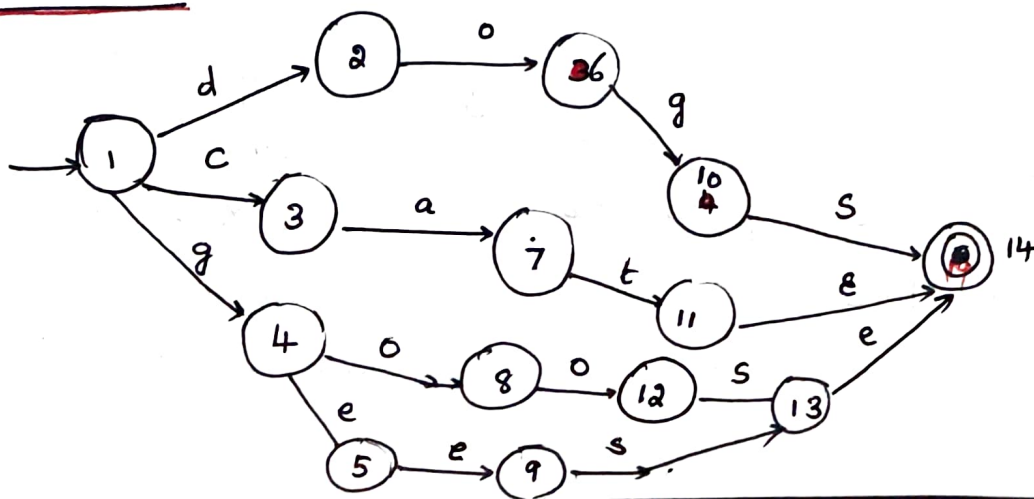
- goose → geese
- Mouse → Mice
- teach → taught
- go → went.

Reg. Noun	Irreg. Plural Noun	Irreg. Singular Noun	Plural for Regular words.
fox	geese	goose	-s
Cat	Sheep	Sheep	
dog	Mice	Mouse	

Std FSA:-

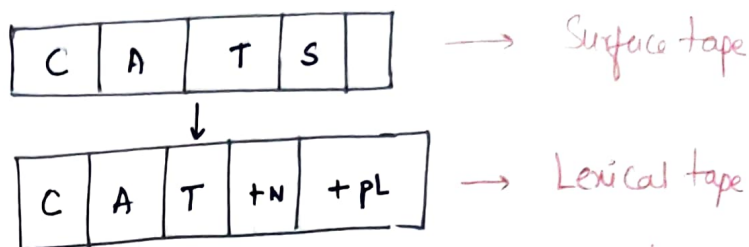


For Certain words



Recognition VS Analysis

- FSA can recognize/accept a string, but they cannot tell its internal structure.
- Machine is required to map/transduce the input string into an output string that encodes its internal structure.
- Finite State Transducers have 2 tapes for i/p and o/p
 - Lexical tape
 - Surface tape.



→ Any one of the tape can be taken as i/p. Surface tape has the root word → Cat, Lexical tape is saying CAT is noun and plural.

Finite State Transducers :-

- A formal Language is not the natural language. but it can be used to model part of natural languages, Such as Phonology, morphology, etc.

• FSTs are FSAs with two tapes

- A FST is 7 tuple = $T = (Q, \Sigma, \Gamma, q_0, F, \delta, \lambda)$ where

Q = Finite set of states

Σ = Finite set of i/p symbols

(Gamma) Γ = Finite set of o/p symbols

q_0 = Initial State

F = Set of Final States

δ = Transition Function mapping

$$\delta = Q \times \Sigma \rightarrow 2^Q$$

λ = o/p Function mapping

$$\lambda: Q \times (\Sigma \cup \epsilon) \rightarrow Q \times (\Gamma \cup \epsilon)$$

Ex:-

$Q = \{0, 1\} \rightarrow \text{States}$

$\Sigma = \{a, b\} \rightarrow \text{inp}$

$\Gamma = \{x, y, z\} \rightarrow \text{olp}$

$\lambda = \{\Sigma \cup \epsilon\} = \{a, b, \epsilon\}$

$\{\Gamma \cup \epsilon\} = \{x, y, z, \epsilon\}$

$$\lambda = \begin{bmatrix} \langle 0, a \rangle & \langle 0, b \rangle & \langle 0, \epsilon \rangle & \langle 0, \epsilon \rangle \\ \langle 1, a \rangle & \langle 1, b \rangle & \langle 1, \epsilon \rangle & \langle 1, \epsilon \rangle \end{bmatrix} \times \begin{bmatrix} \langle 0, x \rangle & \langle 0, y \rangle & \langle 0, z \rangle & \langle 0, \epsilon \rangle \\ \langle 1, x \rangle & \langle 1, y \rangle & \langle 1, z \rangle & \langle 1, \epsilon \rangle \end{bmatrix}$$

$$\delta = \left\{ \begin{array}{l} \langle \langle 0, a \rangle, \langle 0, x \rangle \rangle \\ \langle \langle 0, b \rangle, \langle 0, y \rangle \rangle \\ \langle \langle 0, a \rangle, \langle 1, z \rangle \rangle \end{array} \right\}$$

Ex:-

